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Unit “ 4 : Session “ 3 : SLO “ 1 | Instruction Execution “ Fill in the Blanks

■ I. Fill in the Blanks

1. instruction (fetch) cycle “ fetching an instruction from memory is part of the instruction cycle.
2. Program Counter (PC) “ holds the address of the next instruction to be executed.
3. Instruction Register (IR) “ after fetching, the instruction is placed in the IR for decoding.
4. Control Unit (CU) “ decodes the instruction and generates control signals.
5. Arithmetic Logic Unit (ALU) “ performs arithmetic and logical operations during execution.
6. operand-fetch (decode) stage “ retrieves the operands needed for execution.
7. accumulator “ stores intermediate results during instruction execution.
8. zero flag “ set when the result of an operation is zero.
9. branch (jump) “ a branch instruction alters the normal sequence of execution.
10. write-back (store) “ the phase that writes the result back to memory or registers.